

PROJECT REPORT – Scratch Game: “Food Eating Cat”

1. Project Title

Food Eating Cat – A Scratch Game with Upgrades and Shop System

2. Introduction

“Food Eating Cat” is a Scratch-based interactive game where the player controls a cat that moves around the screen and eats various food items. Each food gives a different score value. The game includes a fully working shop, upgrades, settings, music control, and a win condition. This project demonstrates the use of variables, events, sprite movement, user interface, and logical game progression in Scratch.

3. Objective

The objectives of the game are:

- To design an interactive game using Scratch.
- To apply programming concepts such as movement, variables, conditions, and loops.
- To implement a shop system for buying upgrades.
- To create a settings menu where the player can change controls and music.
- To develop a scoring system where the player wins at 10,000 points.

4. Game Description

In this game, the player controls a cat. The main goal is to move the cat and eat different types of food scattered around the screen. Each food item gives a different score. The score is used to buy upgrades from the shop.

Food Score Table:

Apple – 1

Bread – 2

Banana – 3

Donut – 4

Cheese Puffs – 5

Taco – 6

Watermelon – 7

Muffin – 8

Fruit Bowl – 9

Cake (final upgrade) – 10

5. Features of the Game

1. Cat Movement

- Mouse follow
- Arrow keys
- WASD keys

(Player can switch movement in settings)

2. Scoring & Growth

- Score increases based on food type.
- Cat grows in size up to a limit.

3. Shop System

The shop allows the player to buy:

- Speed upgrades
- Food upgrades
- Growth upgrades

4. Settings Menu

- Turn music ON/OFF
- Change control style
- Show limits of speed & size

5. Win Condition

Player wins the game when score reaches 10,000. A winning message appears.

6. Coding Concepts Used

This project uses Scratch programming concepts:

- Variables (Score, Speed, Size, Money, Control Mode)
- Events (Green flag, button clicks, key press)

- Control blocks (loops, if-else)
- Broadcast messages
- Sensing blocks (touching food, key detection)
- Sound blocks (background music, mute option)

7. Conclusion

“Food Eating Cat” is a complete Scratch game that combines movement, upgrades, shop mechanics, and UI elements. It teaches logical thinking and programming principles, making it ideal for school-level game demonstrations.